

Megan Jacques

(413) 539-8226 | mojacques01@gmail.com | <https://www.linkedin.com/in/meganjacques/> | megjacques01.github.io

EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA

Bachelor of Science in Robotic Engineering, with Distinction, GPA: 3.15/4.0

May 2024

Related Coursework

Unified Robotics III (Manipulation), Unified Robotics IV (Navigation), Software Engineering, Industrial Robotics, AI for Robotics

SKILLS

Programming Languages: MATLAB(Proficient), C (Intermediate), C++ (Intermediate), Python (Intermediate),

HTML(Intermediate), Java(Beginner)

CAD & Engineering Software: NX(Proficient), Teamcenter(Proficient), MATLAB(Proficient),, GITHUB(Proficient), SOLIDWORKS(Proficient)

Robotics & Technical Skills: FDM 3D printing (Proficient), GD&T (Intermediate), AGILE(Intermediate), ROS (Intermediate), JIRA(Intermediate), SLAM(Beginner)

Other Skills: power tools, set design/construction, Microsoft Office Applications(Proficient)

PROJECTS

Initiative Tracker, Personal Project, Hartford, CT

Aug. 2026 - Sept. 2026

- Designed a python project that displays the statuses and current turn of different players during TTRPG gaming.
- Designed and implemented an image state management module that allowed for immediate resetting and display adjustments, reducing error rates in visual output.
- Engineered efficient Python dictionaries that improved TTRPG player status tracking, reducing manual input time

Social Robot (Major Qualifying Project), WPI, Worcester, MA

Sept. 2023 - May 2024

- Designed and modeled 10+ robot parts using **SolidWorks**, optimizing for **FDM 3D Printing**, which sped up the prototype production process
- Collaborated with a team of 6 to integrate facial tracking, personality traits, and movement features
- Delivered weekly progress updates to advisors, managed part orders and a project budget of \$1,000, ensuring all project deadlines were met

Sorting Robot (Unified Robotics III), WPI, Worcester, MA

Aug. 2023 - Oct. 2023

- Developed and applied computer vision and color thresholding algorithms in **MATLAB** to detect and sort 5+ objects with high accuracy, successfully identifying and processing multiple objects
- Calculated **forward and inverse kinematics** for arm movements with 3 degrees of freedom via geometric and algebraic derivation

EXPERIENCE

Drafter, Belcan, Windsor, CT

Dec 2024 - current

- Engineered aerospace drafting packages using Model-Based Definition (MBD) and Geometric Dimensioning and Tolerancing (GD&T) to support design and manufacturing requirements.
- Contributed to continuous improvement of internal workflows and software tools by evaluating engineering software updates, participating in design reviews, troubleshooting bugs, and providing functional feedback.
- Authored technical work-process guides and reference libraries equipped with example part files and instructional slideshows to streamline daily drafting operations.
- Trained and mentored incoming engineering personnel, providing hands-on technical guidance to ensure a smooth transition into the program.

Game Master, Upside Down Escape Games, Mashpee, MA (Seasonal)

May 2023 - Nov. 2024

- Supervised and managed the game flow for 20+ escape room games per week, ensuring seamless experiences for all players
- Supported the design and construction of new props and sets in 2+ escape games, utilizing power tools and 3D printing to enhance the game environment